VDDFormer: A Variable Dependency Discrepancy-Based Transformer for Multivariate Time Series Anomaly Detection

Bo Liu, Lingling Tao, Xiaodan Chen, and Zhijun Li

***Abstract*—The dynamics of multivariate time series (MTS) data are jointly characterized by its nonlinear temporal dependencies and complex variable dependencies, making unsupervised time series anomaly detection a challenging task. Existing methods primarily rely on prediction or reconstruction errors, neglecting the valuable information within the variable dependencies. In this paper, we propose a variable dependency discrepancy-based Transformer (VDDFormer) for unsupervised MTS anomaly detection. VDDFormer comprises a variable correlation encoder, a temporal dependency encoder, and a reconstruction decoder. The variable correlation encoder capitalizes on a variable dependency attention mechanism, which employs self-attention to learn the global variable dependencies; meanwhile, the local variable dependencies are captured by the adaptive correlation matrix. The global and local variable dependencies are then used to compute the variable dependency discrepancy as a new intrinsic property to distinguish between normal and abnormal patterns. By integrating this new discrepancy with the reconstruction error, the model effectively enhances its anomaly differentiation capability. Extensive experiments on five real-world anomaly detection datasets demonstrate that VDDFormer effectively and robustly detects group anomaly patterns by leveraging the variable dependency discrepancy and achieves state-of-the-art performance on four out of the five datasets.**

***Index Terms*—Multivariate time series, anomaly detection, variable dependency discrepancy, variable dependency attention.**

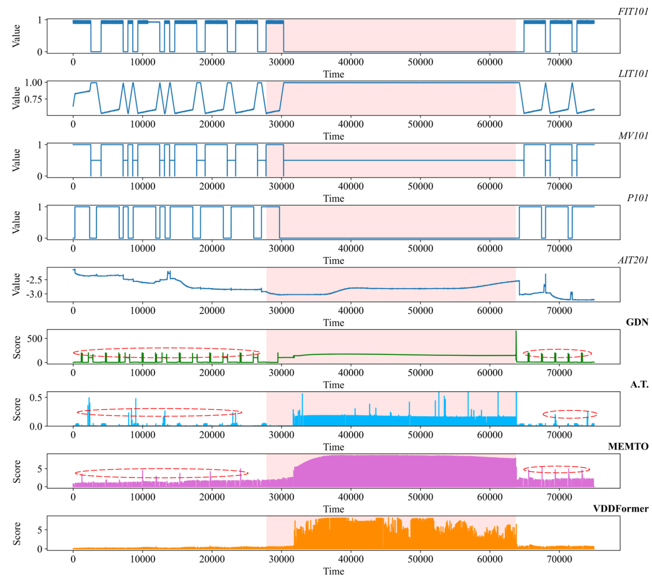

Fig. 1. Example of anomalies in the SWaT dataset and anomaly scores obtained by existing methods. FIT101, LIT101, MV101, P101, and AIT201 are the five sensors in the SWaT dataset. Moments with higher anomaly scores obtained by GDN, Anomaly Transformer (A.T.), MEMTO, and VDDFormer (ours) methods are more likely to be identified as anomalies. The red background interval in the figure represents the abnormal segment. The red dashed circles enclose events that are incorrectly identified as anomalies.

I. Introduction

The proliferation of interconnected devices in modern industrial systems necessitates the use of numerous sensors to monitor their health since anomalies are inevitable and can incur significant economic losses. The continuously recorded sensor data forms a vast collection of multivariate time series. To ensure security and prevent financial repercussions, effectively detecting abnormal time points within the multivariate time series data becomes crucial in modern industrial systems [1], [2], [3]. However, anomalies are often rare and hidden by the sheer volume of normal data in practice. This makes manual labeling of anomalies very expensive and even impractical, rendering supervised anomaly detection techniques unsuitable.

Unsupervised anomaly detection for real-world MTS is challenging due to the intricate temporal and variable dependencies inherent in the data. These dependencies are further complicated by the dynamic variable correlations, where changes in one variable can trigger a cascade of corresponding changes in others. For instance, as illustrated in Fig. 1, the sensors FIT101, LIT101, MV101, and P101 exhibit intricate correlations and synchronized periodic fluctuations. When FIT101 experiences an anomaly, similar abnormalities appear in LIT101, MV101, and P101, demonstrating a *group anomaly pattern* characterized by: during anomalies, long-term and short-term abnormal variable clusters maintain similar dependency patterns (relatively small distribution discrepancy), whereas normal states

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exhibit stable long-term variable dependencies contrasting with dynamic short-term variable interactions (relatively large distributional contrast).

This discrepancy underscores the necessity of examining collective behaviors to reveal group anomaly patterns, while individual and local inspections might miss them.

Spurred by its immense practical utility, a plethora of unsupervised anomaly detection algorithms have been proposed in the literature. Among these, classical statistical learning methods form a notable category, encompassing principal component analysis-based techniques [4], [5], one-class learning-based approaches [6], [7], [8], and density estimation-based methods [9], [10]. These statistical methodologies often depend on manually engineered features and generally fail to account for the temporal dynamics or intricate relationships between variables. As a result, their efficacy in capturing normal patterns within multivariate time series data is compromised. To leverage the powerful representation learning ability of deep neural networks, deep learning-based methods have gained prominence in recent years. These methods involve the use of specifically designed Recurrent Neural Networks (RNNs) [11], [12], [13] that perform self-supervised prediction or reconstruction tasks to discern normal patterns. During the detection phase, reconstruction errors are commonly employed as discriminative criteria, where patterns with substantial reconstruction errors are classified as anomalous.

Despite the advancements in prediction and reconstruction-based anomaly detection methods, a critical oversight in these approaches is the failure to account for the intricate relationships between variables. This oversight can result in sub-optimal performance during reconstruction or prediction, which in turn diminishes the overall effectiveness of the anomaly detection process. To address this shortcoming, several methodologies [14], [15], [16], [17] have emerged that harness the power of Graph Neural Networks (GNNs) or Variational Auto-Encoder networks (VAEs) to model the dependencies or distributions of dependencies among variables. However, these strategies predominantly continue to rely on prediction error or reconstruction error as the benchmarks for identifying anomalies, a criterion that may not suffice in the presence of complex dependencies, as it does not fully exploit the rich information embedded within variable dependencies. Figure 1 highlights that when faced with complex variable interdependencies, the anomaly scores produced by current techniques (such as GDN [14] and MEMTO [18]) may result in erroneous alerts, culminating in the incorrect classification of numerous normal occurrences as anomalies. Recent developments in Transformer-based models have yielded significant progress in time series prediction [19], [20], [21], [22], [23], [24] and anomaly detection [18], [25], largely owing to their wide receptive field. Despite these strides, these approaches have not fully capitalized on the exploitation of variable dependencies, thus constraining their detection capability for group anomalies.

To address the shortcomings of existing methods, in this paper, we propose a VDDFormer developed with a new criterion—variable dependency discrepancy. VDDFormer comprises three components: a variable correlation encoder, a temporal dependency encoder, and a reconstruction decoder. The variable correlation encoder is designed to capture both global and local dependencies, thereby quantifying the variable dependency discrepancy. The temporal dependency encoder is constructed to model the temporal dependencies of MTS data by embedding observations at each time step as tokens and learning their self-attention map to capture temporal dependencies. Finally, the reconstruction decoder is designed to combine the outputs of the above two encoders to reconstruct the data. VDDFormer is developed based on the characteristic that group anomalies exhibit relatively small discrepancies between long-term and short-term variable dependencies, contrasting with relatively large distribution discrepancies in normal states. Specifically, we embed the entire series of each variable within a long time window as individual tokens and compute their self-attention map. This attention map, which can reflect the long-time variable dependencies, is referred to as global dependency. For short-term variable relationships, the dependency weight matrix at each time step is modeled using the adaptive adjacency matrix approach proposed in [26]. This matrix captures instantaneous interactive correlations among variables and is termed local dependency. As both global dependency and local dependency measure the group behavior collectively, they offer us a new criterion, dubbed *variable dependency discrepancy*, to identify group anomaly patterns by quantifying the discrepancy between global dependency and local dependency. The combination of variable dependency discrepancy and reconstruction error establishes the eventual criterion, effectively enhancing the performance of group anomaly detection. In summary, our main contributions are summarized as follows:

- We propose a new criterion for unsupervised MTS anomaly detection, variable dependency discrepancy, which can identify group anomaly patterns that are beyond the detection capability of existing methods.
- We develop a new unsupervised MTS anomaly detection method—VDDFormer—equipped with the proposed variable dependency discrepancy. VDDFormer consists of a variable correlation encoder with variable dependency attention, a temporal dependency encoder for modeling temporal dependencies, and a reconstruction decoder that generates reconstruction output.
- VDDFormer achieves state-of-the-art performance on four out of five public MTS anomaly detection benchmarks, which verifies the effectiveness of our proposed variable dependency discrepancy and anomaly detection method.

The remainder of this paper is organized as follows. In Section II, we briefly review the related work in MTS anomaly detection. The problem statement is presented in Section III. We then give the details of the proposed VDDFormer in Section IV and present the experimental results as well as the corresponding analysis in Section V. Finally, we conclude the paper in Section VI.

II. Related Work

Due to the importance of MTS in various fields, unsupervised MTS anomaly detection is a well-researched area. In this section, we provide an overview of existing approaches, and then discuss the rise of Transformer-based approaches for unsupervised MTS anomaly detection.

A. Unsupervised Multivariate Time Series Anomaly Detection

Unsupervised MTS anomaly detection methods can be broadly categorized into two types of algorithms: classical methods and deep learning-based methods.

Classical methods primarily include one-class learning-based methods and density estimation-based methods. One-class SVM [6] and SVDD [7] are typical one-class learning-based anomaly detection algorithms. Specifically, one-class SVM learns a separating hyperplane to distinguish normal data from anomalies, while SVDD constructs a hypersphere to encapsulate the normal data, both leveraging the concept of support vectors in their modeling. A typical density-based algorithm is LOF [9], which identifies outliers by calculating the local outlier factor. The local outlier factor measures the isolation of an object from its neighbors.

Deep learning methods detect anomalies in MTS by learning normal patterns through neural networks. These models primarily identify anomalies by measuring reconstruction errors (reconstruction-based methods) or prediction errors (prediction-based methods). The reconstruction-based method LSTM-VAE [13] combines Long Short-Term Memory (LSTM) and VAE to respectively capture temporal patterns and reconstruct the expected distribution of normal patterns. Additionally, Generative Adversarial Network (GAN)-based approaches [27], [28], [29] employ adversarial regularization to learn temporal normal patterns for anomaly detection. In contrast to anomaly detection based on reconstruction error, autoregressive models detect anomalies using prediction error. Hundman et al. [11] employ LSTM networks to learn temporal representations and perform anomaly detection by leveraging prediction error. Recently, hybrid methods have been proposed to integrate the strengths of different models for enhanced anomaly detection. For example, Deep SVDD [30] uses deep neural networks to minimize features to separate hyperplanes and identify anomalies. DAGMM [31] and MPPCACD [32] combine deep representation methods with Gaussian Mixture Models (GMMs) to estimate representation density for anomaly detection. Frehner et al. [33] propose an autoencoder-based framework combined with Kernel Density Estimation (KDE) on reconstruction errors to enhance the robustness of time series anomaly detection. ITAD [34] proposes an anomaly detection method based on tensor decomposition and clustering. THOC [35] leverages one-class learning with a dilated RNN to capture multi-scale temporal patterns and employs hierarchical clustering to define one-class targets for anomaly detection.

The aforementioned deep learning methods predominantly model temporal patterns in MTS but neglect the crucial assistance of variable correlations for time series modeling. To address this issue, OmniAnomaly [36] employs a Stochastic Recurrent Neural Network (SRNN) with planar normalization flow to model temporal dependencies in latent variables, enhancing the reconstruction of normal patterns. InterFusion [17] adopts a hierarchical VAE to jointly model temporal dynamics and variable dependencies, ensuring more accurate reconstruction in MTS.

To effectively model relationships between variables, GNNs [37], [38] have been widely applied to anomaly detection tasks. For example, the CoLA algorithm [39] develops a GNN-based contrastive learning framework for anomaly detection in attributed networks. In MTS anomaly detection, GNN-based methods [14], [15] have also been widely adopted. GReLeN [16] leverages VAE for feature representation and integrates GNN to model variable dependencies, thereby enhancing the model's capability to predict normal patterns and to identify anomalies.

B. Transformer-Based Multivariate Time Series Anomaly Detection

Benefiting from the ability of self-attention to capture long-term dependencies, Transformers [40], [41], [42], [43] have achieved remarkable results in MTS analysis. Recently, many Transformer-based models have achieved significant performance improvements in MTS prediction [19], [20], [21], [22], [23], [24], [44]. PatchTST [45] demonstrates effectiveness in prediction through its patching design of Transformer-based model. For the task of MTS anomaly detection, Xu et al. [25] propose the Anomaly Transformer to detect anomalies in MTS. This method combines the association discrepancy of temporal patterns in MTS with the reconstruction error as a detection criterion, achieving effective anomaly detection. Song et al. [18] introduce a memory-guided Transformer architecture employing a reconstruction-based approach to mitigate the over-generalization problem in the anomaly detection task.

Some existing Transformer-based methods [44] embed sub-series of each variable as tokens and apply self-attention to these tokens for time series modeling. While enhancing modeling capabilities through variable relationships, these methods do not adequately explore the discriminative potential of variable correlations for anomaly differentiation. To address this limitation, we propose VDDFormer with a variable dependency attention that simultaneously models global and local dependencies among variables, leveraging their distribution discrepancy as a new intrinsic criterion (termed variable dependency discrepancy) to distinguish normal and abnormal patterns. By integrating this discrepancy with the reconstruction error, VDDFormer achieves robust anomaly detection performance.

III. Problem Statement

Multivariate Time Series: We define the multivariate time series used as input for anomaly detection as $\mathbf{X} \in \mathbb{R}^{T\times N}$, where N variables are observed over T timestamps. The i-th row of $\mathbf{X}$, $\mathbf{x}_i \in \mathbb{R}^N$, corresponds to the observations of all variables at timestamp i. The n-th column of $\mathbf{X}$, denoted as $\mathbf{x}^{(n)} \in \mathbb{R}^T$, represents the observations of the variable n. For a long time series, we use a sliding window of length T to generate fixed-length inputs. Additionally, we use $x_{i,n}$ to denote the observation of the variable n at timestamp i.

Anomaly Detection: The task of multivariate time series anomaly detection is to output a binary vector $\mathbf{y} \in \mathbb{R}^T$ based on

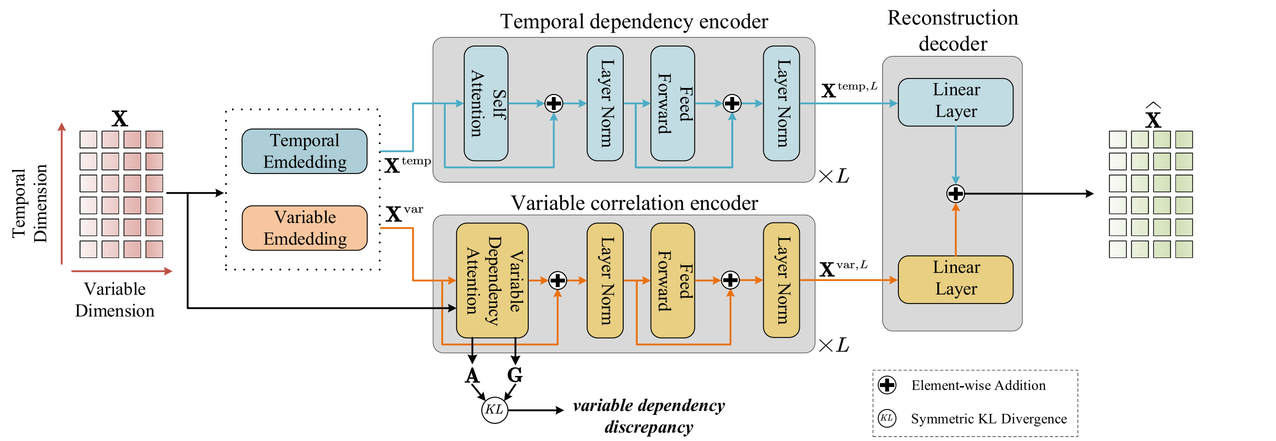


Fig. 2. VDDFormer architecture. VDDFormer consists of a **temporal dependency encoder**, a **variable correlation encoder**, and a **reconstruction decoder**. The temporal dependency encoder learns the temporal dependencies of MTS. Variable correlation encoder utilizes the **variable dependency attention** to embody the *variable dependency discrepancy*, which is the inherent distinguishing property between normal and anomaly based on variable dependency. The reconstruction decoder combines the outputs of the temporal dependency encoder and variable correlation encoder to obtain the final model's reconstruction output.

the given input $\mathbf{X} \in \mathbb{R}^{T\times N}$, where $y_i \in \{0, 1\}$, with 0 indicating a normal time point and 1 denoting an abnormal time point.

IV. Methodology

In this section, we introduce the proposed method VDDFormer. VDDFormer comprises three components: a variable correlation encoder, a temporal dependency encoder, and a reconstruction decoder. The architecture of VDDFormer is shown in Fig. 2. First, we present the variable correlation encoder. Based on the discovery of variable dependency discrepancy, we devise a variable dependency attention mechanism that simultaneously models global and local dependencies. By quantifying the distribution divergence between global and local dependencies, the variable dependency discrepancy is derived and used as a criterion for anomaly detection. In addition, recognizing the importance of modeling temporal dependencies in MTS data, we construct the temporal dependency encoder, which treats the data corresponding to each time step as tokens and models the temporal dependencies of MTS by computing self-attention map. Subsequently, to enable the model to better capture temporal patterns and variable dependencies in MTS data, we design a reconstruction decoder that integrates dual encoder representations, generating temporal-variable reconstructions of MTS. This dual feature fusion mechanism captures joint temporal-variable properties, effectively guiding the model to learn normal patterns in MTS data. Finally, for robust anomaly detection, the combination of reconstruction error and variable dependency discrepancy is utilized as a new anomaly detection criterion.

A. Variable Correlation Encoder

The vanilla single-branch self-attention in Transformers [46] cannot simultaneously capture global dependency and local dependency. To address this limitation, we propose a variable dependency attention mechanism that concurrently models both global and local dependencies to compute variable dependency discrepancy. We construct a variable correlation encoder using this mechanism as the core component. The variable correlation encoder consists of multiple basic modules, each containing a variable dependency attention block and a feed-forward layer. The stacking of multiple modules enables the learning of latent variable dependencies at different levels. Assuming the variable correlation encoder consists of L modules, with length-T input time series denoted as $\mathbf{X} \in \mathbb{R}^{T\times N}$, the operations of the l-th module can be expressed as follows:

$$\begin{aligned}\mathbf{Z}^{\text{var},l} &= \text{LN}\left(\text{VDA}\left(\mathbf{X}^{\text{var},l\text{-}1}, \mathbf{X}\right) + \mathbf{X}^{\text{var},l\text{-}1}\right)\\ \mathbf{X}^{\text{var},l} &= \text{LN}\left(\text{FeedFoward}\left(\mathbf{Z}^{\text{var},l}\right) + \mathbf{Z}^{\text{var},l}\right),\end{aligned} \tag{1}$$

where $\mathbf{X}^{\text{var},l} \in \mathbb{R}^{N\times d_{\text{var}}}, l \in \{1, \cdots, L\}$ denotes the output of the l-th module in the variable correlation encoder with d_{var} channels. The initial input $\mathbf{X}^{\text{var},0} = \text{VariableEmbedding}(\mathbf{X})$ represents the embedded raw series, where $\mathbf{X}^{\text{var},0} \in \mathbb{R}^{N\times d_{\text{var}}}$. Specifically, unlike previous Transformer-based models [18], [25] that embed all variables at the same timestamp as a token, our VariableEmbedding$(\cdot)$ method embeds the entire series of each variable within the sliding window as individual tokens. LN$(\cdot)$ denotes layer normalization as widely adopted in [46]. $\mathbf{Z}^{\text{var},l} \in \mathbb{R}^{N\times d_{\text{var}}}$ is the hidden representation of the l-th module. VDA$(\cdot,\cdot)$ represents the variable dependency attention, which captures global and local dependencies to compute the variable dependency discrepancy. We will elaborate on this attention mechanism in the following content.

Variable Dependency Attention Mechanism: The variable dependency attention mechanism is the core component of the variable correlation encoder, whose pipeline is illustrated in Fig. 3. As aforementioned, the variable dependency discrepancy is computed by measuring the distribution distance between global and local dependencies. To overcome the limitations of vanilla single-branch self-attention, we implement a dual-branch attention mechanism that allows for the concurrent extraction of both global and local dependencies. For local dependency, the adaptive adjacency matrix method [26] is employed to obtain the dependency weights among variables at each time step, followed by the application of the TopK strategy to prune

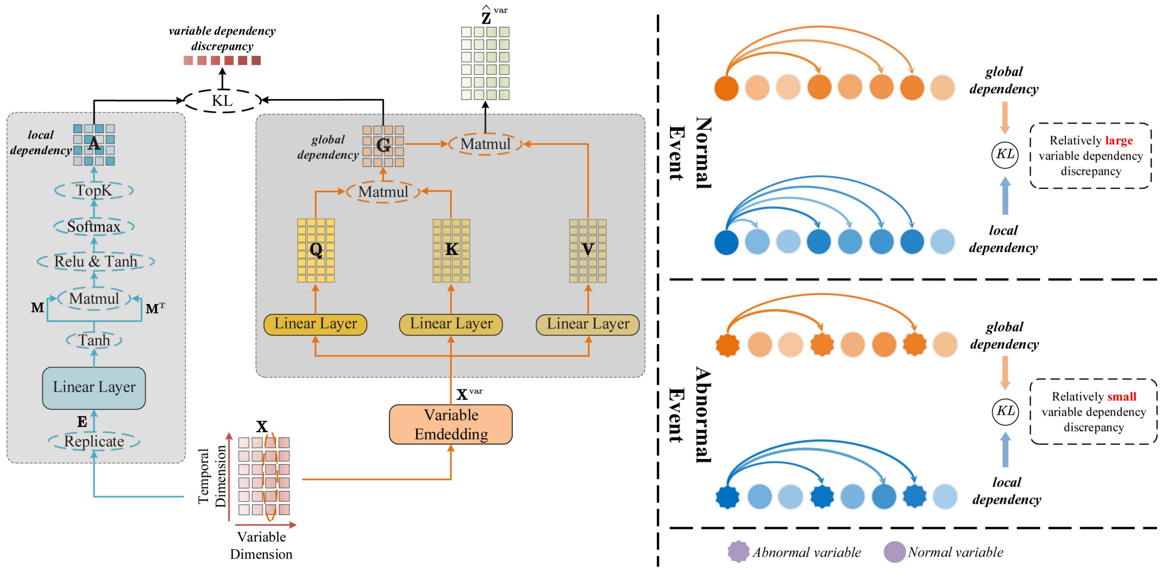


Fig. 3. **Variable dependency attention** mechanism (left) and diagram of ***variable dependency discrepancy*** (right). Variable dependency attention captures the local dependency and global dependency simultaneously. The symmetrized KL divergence between local dependency and global dependency offers us a new criterion, dubbed *variable dependency discrepancy:* In the right figure, the solid lines depict variable dependencies with color intensity indicating weight strength. During anomalies, long-term and short-term abnormal variable clusters exhibit similar dependency patterns (relatively small distribution discrepancy), while normal conditions feature stable long-term dependencies contrasting with dynamic fluctuating short-term dependencies, yielding relatively large distributional contrasts.

negligible weights from the local dependency. Specifically, the local dependency is computed as follows:

$$\mathbf{M} = \mathrm{Tanh}\left(\alpha \mathbf{E}\mathbf{W}_{\mathrm{e}}\right)$$
$$\mathbf{A}^{l} = \mathrm{TopK}\left(\mathrm{Softmax}\left(\mathrm{ReLU}\left(\mathrm{Tanh}\left(\alpha\left(\mathbf{M}\mathbf{M}^{\mathrm{T}}\right)\right)\right)\right)\right), \quad (2)$$

where $\mathbf{E} \in \mathbb{R}^{T\times N\times T}$ is constructed by expanding the input series $\mathbf{X} \in \mathbb{R}^{T\times N}$ with a singleton dimension and replicating it T times to facilitate learning variable dependencies at each timestep. α is a hyperparameter that controls the saturation rate of the $\mathrm{Tanh}(\cdot)$ activation function. The parameter matrix of the linear layer is denoted by $\mathbf{W}_{\mathrm{e}} \in \mathbb{R}^{T\times d_{\mathrm{e}}}$, while the TopK operation retains the top K relation values from the local dependency. The final learned local dependency tensor is represented as $\mathbf{A}^{l} \in \mathbb{R}^{T\times N\times N}$.

In the global dependency branch, the data corresponding to each variable within the sliding window are embedded as tokens. These tokens are fed into a self-attention mechanism to derive the attention map among variables. The weights in the attention map represent long-term dependencies among variables, i.e. the global dependency. The global dependency computation for the l-th module of the variable correlation encoder is formulated as follows:

$$\mathbf{Q} = \mathbf{X}^{\mathrm{var},l\text{-}1}\mathbf{W}_{\mathrm{Q}}^{l}$$
$$\mathbf{K} = \mathbf{X}^{\mathrm{var},l\text{-}1}\mathbf{W}_{\mathrm{K}}^{l}$$
$$\mathbf{V} = \mathbf{X}^{\mathrm{var},l\text{-}1}\mathbf{W}_{\mathrm{V}}^{l}$$
$$\mathbf{G}^{l} = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d_{\mathrm{var}}}}\right)$$
$$\widehat{\mathbf{Z}}^{\mathrm{var},l} = \mathbf{G}^{l}\mathbf{V}, \quad (3)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N\times d_{\mathrm{var}}}$ denote the query, key, and value of the variable dependency attention, respectively. The parameter matrices $\mathbf{W}_{\mathrm{Q}}^{l}, \mathbf{W}_{\mathrm{K}}^{l}, \mathbf{W}_{\mathrm{V}}^{l} \in \mathbb{R}^{d_{\mathrm{var}}\times d_{\mathrm{var}}}$ correspond to $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ in the l-th module. In (3), the discrete weight matrix $\mathbf{G}^{l} \in \mathbb{R}^{N\times N}$ representing global variable dependency is derived via the $\mathrm{Softmax}(\cdot)$ operation. $\widehat{\mathbf{Z}}^{\mathrm{var},l} \in \mathbb{R}^{N\times d_{\mathrm{var}}}$ is the hidden representation of variable dependency attention in the l-th module of the variable correlation encoder.

In summary, the variable dependency attention mechanism is constructed by integrating the two branches, which essentially combines (2) and (3). The function of the mechanism in the l-th module is formalized as:

$$\mathbf{A}^{l}, \mathbf{G}^{l}, \widehat{\mathbf{Z}}^{\mathrm{var},l} = \mathrm{VDA}\left(\mathbf{X}^{\mathrm{var},l\text{-}1}, \mathbf{X}\right), \quad (4)$$

where $\mathrm{VDA}(\cdot,\cdot)$ denotes the overall function of the variable dependency attention, which takes $\mathbf{X}^{\mathrm{var},l\text{-}1}$ and $\mathbf{X}$ as inputs, and produces $\mathbf{A}^{l}$ (local dependency), $\mathbf{G}^{l}$ (global dependency) and $\widehat{\mathbf{Z}}^{\mathrm{var},l}$ (hidden representation) as outputs.

In the implementation of the global dependency branch for the variable dependency attention mechanism, we employ a multi-head attention version with h heads. Computing the variable dependency discrepancy at each time step requires adaptively generating a global dependency matrix for the corresponding time step. To this end, we set the number of attention heads h equal to the input series length T. For the m-th head in the variable dependency attention, the query, key, and value are defined as $\mathbf{Q}_{\mathrm{m}}, \mathbf{K}_{\mathrm{m}}, \mathbf{V}_{\mathrm{m}} \in \mathbb{R}^{N\times \frac{d_{\mathrm{var}}}{h}}$. After computing multi-head attention, the attention maps $\{\mathbf{G}_{m}^{l} \in \mathbb{R}^{N\times N}\}_{1\leq m\leq h}$ are generated. To aggregate information across all heads, these maps are stacked along a new dimension to form a new tensor $\mathbf{G}^{l} \in \mathbb{R}^{h\times N\times N}$. Since h equals T, $\mathbf{G}^{l} \in \mathbb{R}^{T\times N\times N}$. Furthermore, the variable dependency attention concatenates the outputs $\{\widehat{\mathbf{Z}}_m^{\text{var},l} \in \mathbb{R}^{N\times\frac{d_{\text{var}}}{h}}\}_{1\le m\le h}$ of all heads to obtain the final hidden representation of the attention module, denoted as $\widehat{\mathbf{Z}}^{\text{var},l} \in \mathbb{R}^{N\times d_{\text{var}}}$.

Variable Dependency Discrepancy. The local and global dependencies captured by the variable dependency attention mechanism reflect different concentrations on the variable dependency distribution, whereas the prior and series associations captured by Anomaly Transformer [25] reflect different concentrations on the temporal dependency distribution. Inspired by the effectiveness of the symmetrized Kullback-Leibler (KL) divergence in Anomaly Transformer [25] for measuring distribution discrepancy, we adopt it to quantify the discrepancy between local and global dependencies. To amplify the discriminability of the two distributions, we first aggregate and then average the variable dependency discrepancy terms from the L modules of the variable correlation encoder, yielding a more informative metric. The variable dependency discrepancy is computed as:

$$\begin{aligned}\text{VarDepDis}(\mathbf{A},\mathbf{G}) = \Bigg[\frac{1}{L}\sum_{l=1}^{L}\Big(&\text{KL}\left(\mathbf{A}_{i,:,:}^{l} \parallel \mathbf{G}_{i,:,:}^{l}\right)\\ &+\text{KL}\left(\mathbf{G}_{i,:,:}^{l} \parallel \mathbf{A}_{i,:,:}^{l}\right)\Big)\Bigg]_{i=1,\ldots,T}, \quad (5)\end{aligned}$$

where $\text{KL}(\cdot||\cdot)$ denotes the KL divergence between two discrete distributions. The term $\text{VarDepDis}(\mathbf{A},\mathbf{G}) \in \mathbb{R}^{T}$, calculated as the average of multiple module variable dependency discrepancies, reflects a more informative dissimilarity measure. Each element in $\text{VarDepDis}(\mathbf{A},\mathbf{G})$ corresponds to a time step in $\mathbf{X}$. As previously mentioned, global dependencies refer to long-term variable relationships that focus on learning stable inter-variable correlations in extended temporal contexts, thus typically being stable. In normal scenarios, the variable correlations at individual time steps are influenced by data variations and periodic fluctuations, generating local dependencies that capture complex relationships. This results in measurable distribution discrepancies between local and global dependencies under normal conditions. However, during anomalies, long-term and short-term abnormal variable clusters maintain similar dependency patterns. Consequently, the local dependency distribution approximates the global one, leading to relatively small variable dependency discrepancies. This inherent property of the variable dependency discrepancy is leveraged for anomaly detection.

B. Temporal Dependency Encoder

When modeling MTS data, temporal dependencies are equally critical to consider along with variable dependencies. To capture temporal dependency representations, we develop the temporal dependency encoder specifically designed to learn temporal features in MTS data. The encoder embeds all the variables observed at each time step into tokens. These tokens are then fed into a self-attention mechanism to compute an attention map, in which the weights represent temporal dependencies across time steps [25]. Effective learning of temporal dependencies plays a critical role in enhancing model performance. The temporal dependency encoder is constructed by stacking multiple basic modules, each composed of a self-attention block and a feed-forward layer. Through this hierarchical architecture, the encoder further improves the quality of temporal representations. Assuming the temporal dependency encoder consists of L modules, with length-T input time series denoted as $\mathbf{X} \in \mathbb{R}^{T\times N}$, the operations of the l-th module can be expressed as follows:

$$\begin{aligned}\mathbf{Z}^{\text{temp},l} &= \text{LN}\left(\text{SA}\left(\mathbf{X}^{\text{temp},l\text{-}1}\right)+\mathbf{X}^{\text{temp},l\text{-}1}\right)\\ \mathbf{X}^{\text{temp},l} &= \text{LN}\left(\text{SA}\left(\mathbf{Z}^{\text{temp},l}\right)+\mathbf{Z}^{\text{temp},l}\right), \quad (6)\end{aligned}$$

where $\mathbf{X}^{\text{temp},l} \in \mathbb{R}^{T\times d_{\text{temp}}}, l \in \{1,\cdots,L\}$ denotes the output of the l-th module in the temporal dependency encoder with d_{temp} channels. The initial input $\mathbf{X}^{\text{temp},0} = \text{TemporalEmbedding}\ (\mathbf{X})$ represents the embedded raw series, where $\mathbf{X}^{\text{temp},0} \in \mathbb{R}^{T\times d_{\text{temp}}}$. Specifically, $\text{TemporalEmbedding}(\cdot)$ embeds all variables at the same timestamp as a token. $\mathbf{Z}^{\text{temp},l} \in \mathbb{R}^{T\times d_{\text{temp}}}$ represents the hidden representation of the l-th module of the temporal dependency encoder. $\text{LN}(\cdot)$ denotes layer normalization. $\text{SA}(\cdot)$ represents the self-attention.

C. Reconstruction Decoder

To enable the model to better learn the temporal patterns and variable dependencies of normal MTS data, we propose the reconstruction decoder that leverages the outputs of the dual encoders to reconstruct MTS. This dual feature fusion mechanism captures temporal-variable characteristics, effectively guiding the model to learn normal patterns in MTS data. The reconstruction output produced by the reconstruction decoder is formalized as follows:

$$\widehat{\mathbf{X}} = \mathbf{X}^{\text{temp},L}\mathbf{W}_{\text{temp}} + \left(\mathbf{X}^{\text{var},L}\mathbf{W}_{\text{var}}\right)^{\text{T}}, \quad (7)$$

where $\mathbf{X}^{\text{temp},L} \in \mathbb{R}^{T\times d_{\text{temp}}}$ denotes the output of the L-th module (i.e., the final module) in the temporal dependency encoder. $\mathbf{X}^{\text{var},L} \in \mathbb{R}^{N\times d_{\text{var}}}$ represents the output of the L-th module (i.e., the final module) in the variable correlation encoder. $\mathbf{W}_{\text{temp}} \in \mathbb{R}^{d_{\text{temp}}\times N}$, $\mathbf{W}_{\text{var}} \in \mathbb{R}^{d_{\text{var}}\times T}$ denote the parameter matrices. $\mathbf{X}^{\text{temp},L}\mathbf{W}_{\text{temp}} \in \mathbb{R}^{T\times N}$ and $(\mathbf{X}^{\text{var},L}\mathbf{W}_{\text{var}})^{\text{T}} \in \mathbb{R}^{T\times N}$ are combined to generate the final model's reconstruction output $\widehat{\mathbf{X}} \in \mathbb{R}^{T\times N}$.

D. Optimization Strategy and Anomaly Criterion

Optimization Strategy: As an unsupervised anomaly detection task, we utilize reconstruction loss to optimize the model. To enhance the discriminability between normal and abnormal patterns, we introduce the variable dependency discrepancy as an auxiliary loss term. This loss term guides the model to focus more on variable dependencies that facilitate time series reconstruction, thus further sparsifying the global dependency. Consequently, the enlarged distribution discrepancies between local and global dependencies make normal patterns more distinguishable from abnormal patterns with relatively small variable dependency discrepancies. The formula for the loss function is as follows:

$$\begin{aligned}\mathcal{L}\left(\widehat{\mathbf{X}},\mathbf{A},\mathbf{G},\lambda;\mathbf{X}\right) &= \|\mathbf{X}-\widehat{\mathbf{X}}\|_{\text{F}}^{2} - \lambda\\ &\times \|\text{VarDepDis}\left(\mathbf{A},\mathbf{G}\right)\|_{1}, \quad (8)\end{aligned}$$

where $\widehat{\mathbf{X}} \in \mathbb{R}^{T \times N}$ denotes the final output of the model. $\|\cdot\|_F$ and $\|\cdot\|_1$ denote the Frobenius-norm and the 1-norm, respectively. λ is a hyperparameter to trade off the two loss terms. When λ is positive, the optimization process amplifies the variable dependency discrepancy of normal events during training, thereby better distinguishing normal events from abnormal events through the enlarged discrepancy.

Anomaly Criterion: We combine the variable dependency discrepancy with the reconstruction error, using both the temporal-variable representation and the distinguishable variable dependency discrepancy. The final anomaly criterion is shown as follows:

$$\mathrm{AS}(\mathbf{X}) = \mathrm{Softmax}\left(-\mathrm{VarDepDis}(\mathbf{A}, \mathbf{G})\right) \odot \left[\|\mathbf{X}_{i,:} - \widehat{\mathbf{X}}_{i,:}\|_2^2\right]_{i=1,\cdots,T}, \quad (9)$$

where $\mathrm{AS}(\mathbf{X}) \in \mathbb{R}^T$ denotes the anomaly score at each timestamp of $\mathbf{X}$. An anomaly is flagged when the score exceeds a certain threshold. The operator $\odot$ represents element-wise multiplication. Since anomalies exhibit relatively small variable dependency discrepancies, this characteristic distinguishes them from normal patterns. By applying the negation operation to the variable dependency discrepancies and performing the $\mathrm{Softmax}(\cdot)$ computation, then multiplying the resulting values by reconstruction errors (typically large for abnormal events), the anomaly scores become further amplified. Consequently, the integration of the reconstruction error with the variable dependency discrepancy enables more discriminative identification of anomalies, thereby enhancing the model's anomaly detection capability.

Notably, the variable dependency discrepancy captures differences in variable dependencies between short-term and long-term contexts to identify anomalies. In contrast, the association discrepancy in Anomaly Transformer [25] identifies anomalies solely through differences in association scope between normal and abnormal time points but neglects variable correlations. Therefore, the criterion combining reconstruction error with variable dependency discrepancy is more discriminative than the criterion combining reconstruction error with association discrepancy, particularly for detecting group anomalies in MTS.

V. Experiments

This section evaluates the effectiveness of VDDFormer using five public MTS anomaly detection datasets. First, the benchmarks, baselines, experimental setups, and evaluation metrics are described. Subsequently, the experimental results are presented and analyzed, including performance comparisons with existing methods and findings from ablation studies. Finally, systematic investigations are conducted to examine the impacts of hyperparameters in VDDFormer, followed by an analysis of its computational efficiency.

A. Datasets Description

Server Machine Dataset (SMD [36]): This 5-week MTS dataset collects monitoring metrics from 28 servers at a large internet company. Each server forms a sub-dataset split into training and test sets of equal length. The test set labels every timestamp as normal or abnormal. All servers share the same 38 synchronized metrics.

TABLE I
Details of Datasets

Datasets	Application area	Variable dimensions	Training length	Test length
SMD	Server system	38	708405	708420
SMAP	Space system	25	135183	427617
MSL	Space system	55	58317	73729
SWaT	Water Treatment system	51	495000	449919
PSM	Server system	25	132481	87841

Soil Moisture Active Passive Satellite Dataset (SMAP [11]): This dataset originates from telemetry data collected by NASA's Soil Moisture Active Passive satellite. It comprises 55 entities, each characterized by 25 variables. The dataset is partitioned into two subsets: an unlabeled training set and a labeled test set.

Mars Science Laboratory rover Dataset (MSL [11]): This dataset is obtained from telemetry data generated by NASA's Curiosity Mars rover. The MSL dataset comprises 27 entities, each containing 55 variable dimensions. It is split into a training set and a test set. The training set does not contain labels, while the test set contains labels.

Secure Water Treatment Dataset (SWaT [47]): The SWaT dataset originates from experimental data collected on a water treatment operation experiment bench. It comprises 51 variables recorded over 11 days of continuous 24-hour operation. During the first seven days, the system operated under normal conditions, while staged cyber-physical attacks were implemented throughout the remaining four days. Each time step in the final four-day period contains a binary label indicating whether anomalies were present at that specific time point.

Pooled Server Metric Dataset (PSM [48]): This dataset is collected by eBay, with data obtained from multiple internal nodes of the application servers, and consists of 26 variable dimensions. The following Table I provides information about the five public datasets.

B. Baseline Methods

The proposed method is compared with 15 baseline methods. The classical statistical learning approaches include OC-SVM [6], Isolation Forest [49] and LOF [9]). The deep learning methods include reconstruction-based approaches (LSTM-VAE [13], OmniAnomaly [36], InterFusion [17]); prediction-based approaches (LSTM [11], GReLeN [16]); hybrid approaches (Deep SVDD [30], ITAD [34], DAGMM [31], MPPCACD [32], THOC [35]); and Transformer-based methods adhering to the reconstruction paradigm (Anomaly Transformer [25], MEMTO [18]).

- *OC-SVM [6]:* One-Class SVM uses normal data to create a hyperplane. This hyperplane then acts as a boundary to identify abnormal data points.
- *IsolationForest [49]:* Abnormal data are rare and unique, making them more likely to be isolated than normal data. The IsolationForest method employs the isolation forest algorithm to partition the data. After partitioning, nodes closer to the root node in the resulting trees are identified as anomalies.
- *LOF [9]:* LOF determines whether a point is abnormal by comparing its density with its neighboring points.
- *LSTM-VAE [13]:* LSTM-VAE uses LSTM networks for temporal representation and VAE to reconstruct the expected distribution of normal patterns. It detects anomalies through large reconstruction errors caused by the model's inability to reconstruct anomalies.
- *OmniAnomaly [36]:* OmniAnomaly applies stochastic recurrent neural networks and planar normalization flow techniques to capture the temporal dependencies among random variables in space. Then it discriminates anomalies by reconstructing probabilities.
- *InterFusion [17]:* InterFusion uses a hierarchical VAE to jointly model temporal patterns and variable dependencies, ensuring more accurate reconstruction in MTS.
- *Deep SVDD [30]:* Deep SVDD uses a neural network to minimize the feature space of samples and defines a hypersphere. It then uses the distance between sample points and the hypersphere's center to identify anomalies.
- *ITAD [34]:* ITAD first constructs a third-order tensor for the entire dataset. It then decomposes the tensor into component matrices and performs clustering analysis. Anomalies are identified by calculating the distance between each sample and its cluster centroid.
- *DAGMM [31]:* DAGMM combines an autoencoder with a Gaussian Mixture Model to generate low-dimensional representations of data points, and uses the reconstruction error to identify anomalies.
- *MPPCACD [32]:* MPPCACD detects anomalies using a method based on probabilistic dimensionality reduction and clustering.
- *THOC [35]:* THOC uses dilated recurrent neural networks to capture multi-scale temporal patterns. It then detects anomalies by defining a single-class object through hierarchical clustering.
- *LSTM [11]:* The method employs an LSTM network for anomaly detection, combined with a complementary nonparametric unsupervised thresholding strategy to determine anomaly criterion.
- *GReLeN [16]:* GReLeN uses VAE for feature extraction and combines GNN with a random graph relation learning strategy to capture dependencies between variables, enabling effective anomaly detection.
- *Anomaly Transformer [25]:* Anomaly Transformer exploits the inherent distinguishability between normal and anomaly in the temporal dimension, termed as association discrepancy. Then, this method combines the association discrepancy with the reconstruction error as a detection criterion, achieving effective anomaly detection.
- *MEMTO [18]:* MEMTO uses a memory-guided Transformer based on the reconstruction paradigm to learn normal patterns. This approach restricts the encoder from capturing anomaly features, making abnormal data harder to reconstruct.

C. Experimental Setup

Experimental Setup: In our experiments, the model inputs are sub-series extracted using non-overlapping sliding windows with a fixed length T of 100 across all datasets. A time point is identified as anomalous if its anomaly score within a sub-series exceeds the threshold δ. Following the threshold selection criterion from Anomaly Transformer [25], we set δ based on the top-r percent of anomaly scores in the validation set, specifically $r = 0.3\%$ for SWaT and SMD, 0.5% for PSM, and 1% for the other datasets. Additionally, we use the adjustment method widely adopted in anomaly detection [25], [35], [36], [50] to evaluate the experimental results. Both the temporal dependency encoder and variable correlation encoder consist of three modules. We set the number of hidden channels d_{temp} in the temporal dependency encoder to 512 and configure the number of attention heads to 8. The hidden channels d_{var} in the variable correlation encoder is set to 800, and the number of heads h is set to 100. The output feature dimension d_{e} in the linear layer of (2) is set to 64. The hyperparameter λ in (8) is set to 3 to trade off the terms of reconstruction loss and variable dependency discrepancy. The parameter K in TopK is set to half the number of variables. The Adam [51] optimizer is used in the experiments with an initial learning rate of 10^{-4}. The batch size in the experiment is set to 128, and the early stopping strategy is adopted. We implement VDDFormer with PyTorch [52] and conduct all experiments using a single TESLA V100S 32 GB GPU.

Evaluation Metrics: In this paper, we utilize 3 commonly used evaluation metrics for anomaly detection: Precision (P), Recall (R), and F1 score (F1). These metrics are used to evaluate the performance of the VDDFormer. The formula for calculating the F1 score is as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{10}$$

D. Main Results

Following the experimental setup previously introduced, we conduct comparative experiments between VDDFormer and 15 baseline models across five public datasets. The experimental results are summarized in Table II. We reproduce the results of GReLeN, Anomaly Transformer, and MEMTO, while adopting the performance metrics reported in [25] for other baselines.

As shown in Table II, we first observe that deep learning-based anomaly detection methods exhibit significantly better performance compared to classical approaches. This discrepancy arises because classical methods cannot explicitly model the temporal dependencies and variable correlations inherent

TABLE II
COMPARISON WITH OTHER BASELINES

Model	SMD			MSL			SMAP			SWaT			PSM		
	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
OC-SVM	44.34	76.72	56.19	59.78	86.87	70.82	53.85	59.07	56.34	45.39	49.22	47.23	62.75	80.89	70.67
IsolationForest	42.31	73.29	53.64	53.94	86.54	66.45	52.39	59.07	55.53	49.29	44.95	47.02	76.09	92.45	83.48
LOF	56.34	39.86	46.68	47.72	85.25	61.18	58.93	56.33	57.60	72.15	65.43	68.62	57.89	90.49	70.61
Deep SVDD	78.54	79.67	79.10	91.92	76.63	83.58	89.93	56.02	69.04	80.42	84.45	82.39	95.41	86.49	90.73
ITAD	86.22	73.71	79.48	69.44	84.09	76.07	82.42	66.89	73.85	63.13	52.08	57.08	72.80	64.02	68.13
DAGMM	67.30	49.89	57.30	89.60	63.93	74.62	86.45	56.73	68.51	89.92	57.84	70.40	93.49	70.03	80.08
MMPCACD	71.20	79.28	75.02	81.42	61.31	69.95	88.61	75.84	81.73	82.52	68.29	74.73	76.26	78.35	77.29
THOC	79.76	90.95	84.99	88.45	90.97	89.69	92.06	89.34	90.68	83.94	86.36	85.13	88.14	90.99	89.54
LSTM	78.55	85.28	81.78	85.45	82.50	83.95	89.41	78.13	83.39	86.15	83.27	84.69	76.93	89.64	82.80
LSTM-VAE	75.76	90.08	82.30	85.49	79.94	82.62	92.20	67.75	78.10	76.00	89.50	82.20	73.62	89.92	80.96
OmniAnomaly	83.68	86.82	85.22	89.02	86.37	87.67	92.49	81.99	86.92	81.42	84.30	82.83	88.39	74.46	80.83
InterFusion	87.02	85.43	86.22	81.28	92.70	86.62	89.77	88.52	89.14	80.59	85.58	83.01	83.61	83.45	83.52
GReLeN	93.37	69.69	79.81	69.57	68.29	68.92	50.51	90.48	64.83	98.69	72.78	83.78	92.87	90.46	91.65
Anomaly Transformer	88.84	92.76	90.76	91.69	93.64	92.65	93.59	99.24	96.33	91.55	96.73	94.07	97.26	97.42	97.34
MEMTO	91.37	91.21	**91.29**	90.30	93.43	91.84	93.00	99.71	96.24	91.85	96.75	94.24	95.69	99.41	97.52
VDDFormer	91.47	88.25	89.83	91.18	98.40	**94.66**	94.94	98.86	**96.86**	91.05	99.57	**95.12**	97.71	98.38	**98.04**

Higher values of the three evaluation metrics P, R, and F1 represent better anomaly detection performance.

in MTS data. Consequently, classical methods do not capture complex temporal variations and dynamic dependencies among variables, limiting their effectiveness in addressing the intricate patterns of real-world MTS anomalies. Additionally, deep learning-based approaches that explicitly model variable dependencies consistently outperform those neglecting such interactions. For instance, methods such as ITAD, DAGMM, MPPCACD, LSTM, and LSTM-VAE, which disregard variable dependencies, yield suboptimal performance in comparison. These results demonstrate that capturing dependencies among variables is critical for effective time series modeling and should not be omitted in anomaly detection frameworks.

The results in Table II further demonstrate the advantage of Transformer-based models for MTS anomaly detection, such as Anomaly Transformer, MEMTO, and our proposed VDDFormer. These methods validate the effectiveness of the Transformer architecture in dynamically modeling temporal patterns and variable dependencies. Notably, VDDFormer achieves state-of-the-art performance on four benchmark datasets (SMAP, MSL, SWaT, and PSM), while maintaining comparable results on SMD.

To clarify the technical distinctions, we elaborate on the key differences between VDDFormer and Anomaly Transformer. In principle, Anomaly Transformer focuses on temporal feature modeling. It employs an attention divergence termed association discrepancy to distinguish anomalies, where abnormal patterns primarily attend to adjacent timestamps while normal patterns exhibit holistic temporal dependencies. The association discrepancy is implemented through a dual-branch Anomaly-Attention mechanism. The prior-association branch computes temporal priors using a learnable Gaussian kernel based on relative time distances. The series-association branch embeds multivariate observations at each timestamp into tokens and learns series dependencies through self-attention. The association discrepancy is quantified by measuring the symmetrized KL divergence between the attention weight distributions of both branches, enabling intrinsic discrimination between normal and abnormal patterns. However, Anomaly Transformer design does not explicitly address the intricate variable dependencies inherent in MTS. Our VDDFormer employs a variable dependency attention mechanism to capture the aforementioned variable dependency discrepancy for anomaly detection. Inspired by the Anomaly-Attention architecture, the variable dependency attention mechanism comprises two branches: the local dependency branch captures variable dependencies at each timestamp via an adaptive adjacency matrix method; the global dependency branch embeds subseries of each variable as tokens to learn long-term variable dependencies through self-attention. The variable dependency discrepancy is quantified by computing the statistical distance between global and local dependencies using symmetrized KL divergence. MTS encompass both dynamic temporal patterns and complex inter-variable relationships, rendering purely temporal modeling prone to misjudgment due to interference. As illustrated in Fig. 1, Anomaly Transformer generates false alarms during normal periods, while VDDFormer effectively mitigates this issue through its proposed variable dependency discrepancy criterion.

E. Ablation Analysis

Model Architecture and Discriminant Criteria: To impartially evaluate the effectiveness and contributions of both

TABLE III
ABLATION STUDY ON MODEL ARCHITECTURE AND ANOMALY CRITERION

Architecture	Anomaly Criterion		Dataset					Avg F1 (%)
	Recon	VarDepDis	SMD	MSL	SMAP	SWaT	PSM	
Temporal Dependency Encoder	✓	–	79.47	72.99	73.18	76.36	78.60	76.12
Variable Correlation Encoder	✓	✗	73.54	83.86	67.73	78.69	79.00	76.56
	✗	✓	84.61	93.53	96.26	94.72	97.37	93.30
	✓	✓	86.35	94.27	96.39	95.00	97.49	93.90
VDDFormer	✓	✗	78.92	72.80	78.64	76.39	78.57	77.06
	✗	✓	85.42	94.25	96.21	95.03	97.81	93.74
	✓	✓	**89.83**	**94.66**	**96.86**	**95.12**	**98.04**	**94.90**

Recon and VarDepDis denote the reconstruction error and variable dependency discrepancy respectively.

VDDFormer's components and its discrimination criteria, comprehensive ablation studies are designed. First, a baseline is established by exclusively employing the temporal dependency encoder with reconstruction error as the anomaly criterion. This configuration evaluates anomaly detection capability in the absence of variable correlation encoder. Then, the temporal dependency encoder is removed while retaining only the variable correlation encoder. Three distinct criteria are examined: the reconstruction error, the variable dependency discrepancy, and the composite anomaly criterion proposed in (9). This setting investigates model performance without temporal dependency modeling. Finally, the complete model is evaluated using three distinct criteria: reconstruction error, variable dependency discrepancy, and the proposed composite anomaly criterion. This systematic approach enables thorough validation of the contributions and effectiveness of both the components and the discrimination criteria.

The ablation study results for the three configurations are presented in Table III, yielding the following conclusions. First, the proposed composite anomaly discriminant criterion (combining reconstruction error and variable dependency discrepancy) significantly outperforms the conventional reconstruction error-based criterion. As evidenced by the variable correlation encoder and the VDDFormer results, the composite criterion improves the average F1 score by 17.34% (76.56%→93.90%) and 17.84% (77.06%→94.90%), respectively, compared to using reconstruction error alone. These improvements demonstrate the critical role of jointly considering the reconstruction error and the variable dependency discrepancy for effective anomaly detection. Second, the experimental results demonstrate that the temporal dependency encoder achieves an average F1 score of 76.12%. This reveals the limitations of the model in anomaly detection when it focuses solely on temporal pattern learning while neglecting variable dependency discrepancies. Third, when solely utilizing variable dependency discrepancy as the discrimination criterion, both the variable correlation encoder and VDDFormer deliver superior performance (93.30% and 93.74%, respectively), demonstrating the inherent discriminative capability of the variable dependency discrepancy to distinguish normal from abnormal patterns. Finally, under the composite anomaly criterion, VDDFormer achieves an average F1 score improvement of 1% (93.90%→94.90%) compared to the variable correlation encoder. This outcome substantiates the effectiveness and necessity of the temporal dependency encoder in capturing temporal dependencies within MTS.

Statistical Distance Methods: To investigate the impact of the variable dependency discrepancy computed by diverse statistical distance methods on anomaly detection performance, we conducted comparative experiments using the following widely adopted statistical approaches:

- L2 Distance (L2).
- Cross-Entropy (CE).
- Jensen-Shannon Divergence (JSD).
- Symmetrized Kullback-Leibler (KL) Divergence (Ours).

The experimental results in Table IV demonstrate that the symmetrized KL divergence method achieves the best anomaly detection performance across all five datasets with an average F1 score of 94.90%. While the L2 distance method (90.19% average F1 score), CE method (92.34%), and JSD method (90.79%) exhibit comparable performance across the five datasets, they still show a noticeable gap compared to the symmetrized KL divergence method. Based on these experimental results, the symmetrized KL divergence is adopted to quantify variable dependency discrepancies.

Multi-Module Quantization: During anomaly detection, we calculate the average of variable dependency discrepancies across all modules and integrate it with the reconstruction error to derive the final anomaly score. The multi-module quantification is validated through comparison with all single-module counterparts, where the anomaly criterion is defined by the integration of each module's dependency discrepancy and reconstruction error. As demonstrated in Table V, the multi-module approach outperforms all single-module counterparts. The results not only confirm the efficacy of VDDFormer's multi-module stacking design, but also reveal that multi-module quantification captures more informative variable dependency discrepancies.

F. Analysis of Hyperparameters and Computational Efficiency

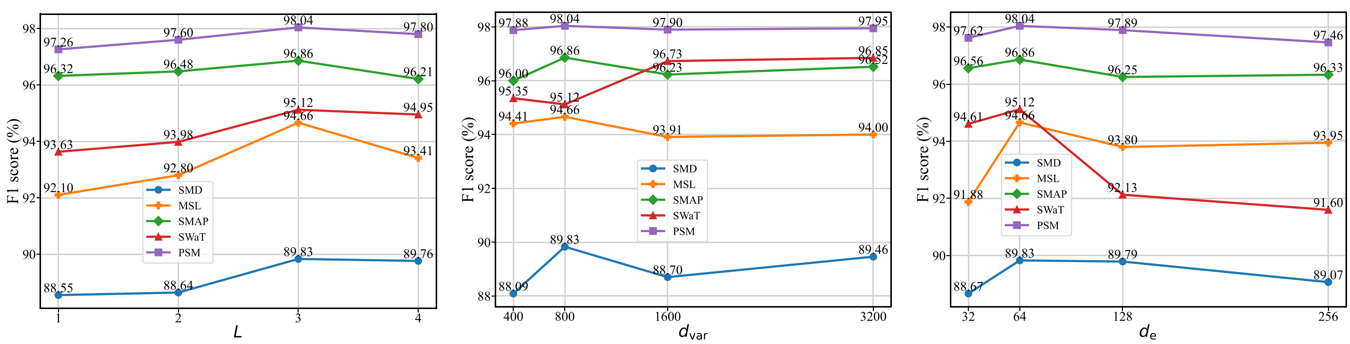
Analysis of Hyperparameters: The hyperparameters of the model are experimentally analyzed. First, the effect of the number of modules on detection performance is studied by varying the module count L from 1 to 4. The derived variable dependency discrepancies are averaged and integrated with the reconstruction error to calculate the anomaly scores for evaluation. As illustrated in Fig. 4(a), three modules achieve optimal performance across all datasets, thereby being adopted as the final configuration. Next, the impact of hidden channels d_{var} is investigated by testing values of 400, 800, 1600, and 3200. The results in Fig. 4(b) show that a channel number of 800 yields the best overall performance, serving as the default setting. The output feature dimension d_{e} in the linear layer of (2) is further analyzed, and Fig. 4(c) shows that it achieves optimal performance when set to 64. These parametric studies reveal limited sensitivity to parameter variations, demonstrating the robustness of VDDFormer in practical applications.

TABLE IV
MODEL PERFORMANCE UNDER DIFFERENT DEFINITIONS OF VARIABLE DEPENDENCY DISCREPANCY

Dataset	SMD			MSL			SMAP			SWaT			PSM			Avg F1 (%)
Metric	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	
L2	91.23	82.50	86.64	86.86	94.57	90.55	94.58	90.75	92.62	81.42	91.75	86.28	90.91	99.19	94.87	90.19
CE	94.70	81.24	87.46	89.35	91.32	90.33	94.17	92.92	93.54	89.01	97.03	92.85	96.91	98.11	97.51	92.34
JSD	92.25	84.40	88.15	90.97	86.51	88.68	89.87	98.66	94.06	91.44	85.68	88.47	90.88	98.58	94.58	90.79
Ours	91.47	88.25	**89.83**	91.18	98.40	**94.66**	94.94	98.86	**96.86**	91.05	99.57	**95.12**	97.71	98.38	**98.04**	**94.90**

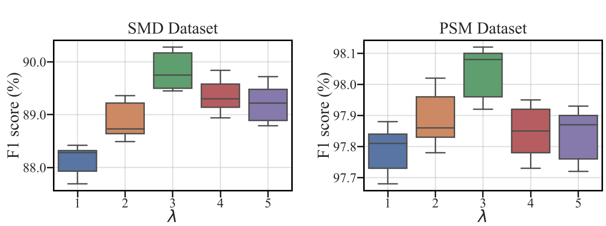
TABLE V
ANOMALY DETECTION PERFORMANCE UNDER DIFFERENT SELECTIONS OF MODULES FOR VARIABLE DEPENDENCY DISCREPANCY

Dataset	SMD			MSL			SMAP			SWaT			PSM			Avg F1 (%)
Metric	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	
Module1	86.70	91.90	89.22	91.97	96.43	94.15	93.95	99.11	96.46	90.06	100.00	94.77	96.82	98.32	97.56	94.43
Module2	87.22	91.58	89.35	86.44	96.28	91.09	93.58	88.41	90.92	70.40	99.57	82.48	96.30	98.33	97.30	90.23
Module3	90.93	84.39	87.54	82.90	89.97	86.29	90.29	80.77	85.26	70.23	97.54	81.66	96.05	98.33	97.17	87.58
Multiple-Module	91.47	88.25	**89.83**	91.18	98.40	**94.66**	94.94	98.86	**96.86**	91.05	99.57	**95.12**	97.71	98.38	**98.04**	**94.90**



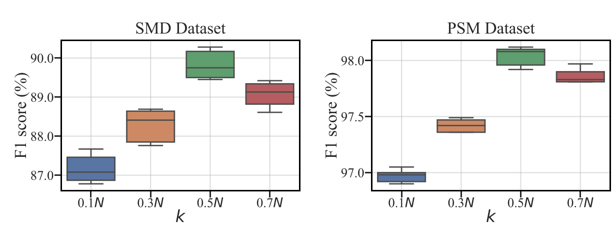
(a) Number of Modules L (b) Hidden Channels d_{var} (c) Linear Layer Output Feature Dimension d_{e}

Fig. 4. Sensitivity analysis of model hyperparameters. (a) Impact of the number of modules L in VDDFormer on detection performance. (b) Impact of the hidden channels d_{var} on detection performance. (c) Impact of the output feature dimension d_{e} of the linear layer on detection performance.



(a) Loss Weight (b) Loss Weight

Fig. 5. Experimental results of the loss weight λ. (a) Performance of different loss weight values on the SMD dataset. (b) Performance of different loss weight values on the PSM dataset.



(a) K Values in TopK Strategy (b) K Values in TopK Strategy

Fig. 6. Experimental results of the K in the TopK strategy. (a) Performance of different K values in the TopK strategy on the SMD dataset. (b) Performance of different K values in the TopK strategy on the PSM dataset. The horizontal axis labels (e.g., $0.1N, 0.3N$) indicate the fraction of the total variables N.

To investigate the impact of loss weight λ and TopK parameter K on detection results, experiments are conducted on the SMD and PSM datasets. The experimental results are presented in Figs. 5 and 6. Optimal detection performance occurs when λ is equal to three, as shown in Fig. 5(a) and (b). Deviating from this value reduces performance, confirming that balanced integration of reconstruction loss and variable dependency discrepancy strengthens normal-abnormal separability for improved detection. Fig. 6(a) and (b) demonstrate that setting K at half the total number of variables achieves superior results. Performance declines as K decreases, indicating that excessive information loss in local dependency weakens the distinction between normal and abnormal patterns, thereby reducing performance.

Computational Efficiency. The time complexity of VDDFormer is analyzed by examining its three main components. The temporal dependency encoder exhibits a time complexity of $\mathcal{O}(Td_{\text{temp}}^2)$, while the variable correlation encoder has a time complexity of $\mathcal{O}(Nd_{\text{var}}^2 + T^2Nd_{\text{e}})$. Meanwhile, the reconstruction decoder exhibits a time complexity of $\mathcal{O}(Td_{\text{temp}}N + Td_{\text{var}}N)$. Under the experimental parameter settings, the inequality $(Td_{\text{temp}}N + Td_{\text{var}}N) \ll Td_{\text{temp}}^2 < Nd_{\text{var}}^2$ holds. Furthermore, given that T^2Nd_{e} equals Nd_{var}^2 (with $T = 100$, $d_{\text{e}} = 64$, $d_{\text{var}} = 800$), the overall time complexity is $\mathcal{O}(Nd_{\text{var}}^2)$. For comparison, the time complexity of Anomaly Transformer is $\mathcal{O}(Td_{\text{temp}}^2)$. Under the given parameter configuration, VDDFormer achieves the same asymptotic order of time complexity as Anomaly Transformer. Additionally, inference time comparisons are performed on the SWaT dataset. For the same 100 samples, VDDFormer achieves an average inference time of 7.52 ms per sample, compared to 4.26 ms per sample for Anomaly Transformer. Despite being marginally slower, VDDFormer maintains computational practicality for real-time anomaly detection scenarios.

VI. Conclusion

This paper proposes a new criterion for MTS anomaly detection, variable dependency discrepancy, which can identify group anomaly patterns that are beyond the detection ability of existing methods. Based on the proposed variable dependency discrepancy, we have developed an unsupervised MTS anomaly detection method, dubbed VDDFormer. VDDFormer consists of a variable correlation encoder with variable dependency attention mechanism, a temporal dependency encoder to model temporal dependencies, and a reconstruction decoder that generates reconstruction output. The integration of reconstruction error and variable dependency discrepancy as a joint discriminative criterion effectively improves anomaly detection performance.

The limitation of VDDFormer lies in its dual-encoder design, which increases the complexity of the model. Specifically, the temporal dependency encoder models temporal dependencies, while the variable correlation encoder captures both global and local dependencies to quantify variable dependency discrepancies. To address this limitation, future work will focus on developing a unified encoder capable of simultaneously modeling temporal dependencies, capturing both global and local dependencies to quantify variable dependency discrepancies. This unified encoder will simplify the model architecture and accelerate anomaly detection.

References

[1] W. Haider, J. Hu, Y. Xie, X. Yu, and Q. Wu, “Detecting anomalous behavior in cloud servers by nested-arc hidden SEMI-Markov model with state summarization,” *IEEE Trans. Big Data*, vol. 5, no. 3, pp. 305–316, Sep. 2019.

[2] N. Moustafa, J. Slay, and G. Creech, “Novel geometric area analysis technique for anomaly detection using trapezoidal area estimation on large-scale networks,” *IEEE Trans. Big Data*, vol. 5, no. 4, pp. 481–494, Dec. 2019.

[3] C. Yang, Z. Du, X. Meng, X. Zhang, X. Hao, and D. A. Bader, “Anomaly detection in catalog streams,” *IEEE Trans. Big Data*, vol. 9, no. 1, pp. 294–311, Feb. 2023.

[4] R. Paffenroth, P. Du Toit, R. Nong, L. Scharf, A. P. Jayasumana, and V. Bandara, “Space-time signal processing for distributed pattern detection in sensor networks,” *IEEE J. Sel. Top. Signal Process.*, vol. 7, no. 1, pp. 38–49, Feb. 2013.

[5] H. Hoffmann, “Kernel PCA for novelty detection,” *Pattern Recognit.*, vol. 40, no. 3, pp. 863–874, 2007.

[6] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, “Estimating the support of a high-dimensional distribution,” *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, 2001.

[7] D. M. Tax and R. P. Duin, “Support vector data description,” *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, 2004.

[8] C. Huang, G. Min, Y. Wu, Y. Ying, K. Pei, and Z. Xiang, “Time series anomaly detection for trustworthy services in cloud computing systems,” *IEEE Trans. Big Data*, vol. 8, no. 1, pp. 60–72, Feb. 2022.

[9] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.

[10] J. Zhao, F. Deng, J. Zhu, and J. Chen, “Searching density-increasing path to local density peaks for unsupervised anomaly detection,” *IEEE Trans. Big Data*, vol. 9, no. 4, pp. 1198–1209, Aug. 2023.

[11] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, “Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2018, pp. 387–395.

[12] C. Zhang et al., “A deep neural network for unsupervised anomaly detection and diagnosis in multivariate time series data,” in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 1409–1416.

[13] D. Park, Y. Hoshi, and C. C. Kemp, “A multimodal anomaly detector for robot-assisted feeding using an LSTM-based variational autoencoder,” *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 1544–1551, Jul. 2018.

[14] A. Deng and B. Hooi, “Graph neural network-based anomaly detection in multivariate time series,” in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 4027–4035.

[15] H. Zhao et al., “Multivariate time-series anomaly detection via graph attention network,” in *Proc. IEEE Int. Conf. Data Mining*, 2020, pp. 841–850.

[16] W. Zhang, C. Zhang, and F. Tsung, “GRELEN: Multivariate time series anomaly detection from the perspective of graph relational learning,” in *Proc. Int. Joint Conf. Artif. Intell.*, 2022, pp. 2390–2397.

[17] Z. Li et al., “Multivariate time series anomaly detection and interpretation using hierarchical inter-metric and temporal embedding,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2021, pp. 3220–3230.

[18] S. Junho, K. Keonwoo, O. Jeonglyul, and C. Sungzoon, “MEMTO: Memory-guided transformer for multivariate time series anomaly detection,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 57947–57963.

[19] H. Wu, J. Xu, J. Wang, and M. Long, “AutoFormer: Decomposition transformers with auto-correlation for long-term series forecasting,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 22 419–22 430.

[20] H. Zhou et al., “Informer: Beyond efficient transformer for long sequence time-series forecasting,” in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 11 106–11 115.

[21] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, and R. Jin, “FedFormer: Frequency enhanced decomposed transformer for long-term series forecasting,” in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 27 268–27 286.

[22] S. Li et al., “Enhancing the locality and breaking the memory bottleneck of transformer on time series forecasting,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 5244–5254.

[23] S. Liu et al., “Pyraformer: Low-complexity pyramidal attention for long-range time series modeling and forecasting,” in *Proc. Int. Conf. Learn. Representations*, 2022.

[24] D. Du, B. Su, and Z. Wei, “Preformer: Predictive transformer with multi-scale segment-wise correlations for long-term time series forecasting,” in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2023, pp. 1–5.

[25] J. Xu, H. Wu, J. Wang, and M. Long, “Anomaly transformer: Time series anomaly detection with association discrepancy,” in *Proc. Int. Conf. Learn. Representations*, 2022.

[26] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang, “Connecting the dots: Multivariate time series forecasting with graph neural networks,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2020, pp. 753–763.

[27] D. Li, D. Chen, B. Jin, L. Shi, J. Goh, and S.-K. Ng, “MAD-GAN: Multivariate anomaly detection for time series data with generative adversarial networks,” in *Proc. Int. Conf. Artif. Neural Netw.*, 2019, pp. 703–716.
[28] B. Zhou, S. Liu, B. Hooi, X. Cheng, and J. Ye, “BeatGAN: Anomalous rhythm detection using adversarially generated time series,” in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, pp. 4433–4439.
[29] J. Audibert, P. Michiardi, F. Guyard, S. Marti, and M. A. Zuluaga, “USAD: Unsupervised anomaly detection on multivariate time series,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2020, pp. 3395–3404.
[30] L. Ruff et al., “Deep one-class classification,” in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4393–4402.
[31] B. Zong et al., “Deep autoencoding gaussian mixture model for unsupervised anomaly detection,” in *Proc. Int. Conf. Learn. Representations*, 2018.
[32] T. Yairi, N. Takeishi, T. Oda, Y. Nakajima, N. Nishimura, and N. Takata, “A data-driven health monitoring method for satellite housekeeping data based on probabilistic clustering and dimensionality reduction,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 3, pp. 1384–1401, Jun. 2017.
[33] R. Frehner, K. Wu, A. Sim, J. Kim, and K. Stockinger, “Detecting anomalies in time series using kernel density approaches,” *IEEE Access*, vol. 12, pp. 33420–33439, 2024.
[34] Y. Shin et al., “ITAD: Integrative tensor-based anomaly detection system for reducing false positives of satellite systems,” in *Proc. Int. Conf. Inf. Knowl. Manage.*, 2020, pp. 2733–2740.
[35] L. Shen, Z. Li, and J. Kwok, “Timeseries anomaly detection using temporal hierarchical one-class network,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 13 016–13 026.
[36] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, “Robust anomaly detection for multivariate time series through stochastic recurrent neural network,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2019, pp. 2828–2837.
[37] K. Ding, Q. Zhou, H. Tong, and H. Liu, “Few-shot network anomaly detection via cross-network meta-learning,” in *Proc. Int. World Wide Web Conf.*, 2021, pp. 2448–2456.
[38] M. Mesgaran and A. B. Hamza, “Graph fairing convolutional networks for anomaly detection,” *Pattern Recognit.*, vol. 145, 2024, Art. no. 109960.
[39] Y. Liu, Z. Li, S. Pan, C. Gong, C. Zhou, and G. Karypis, “Anomaly detection on attributed networks via contrastive self-supervised learning,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2378–2392, Jun. 2022.
[40] Z. Zhang et al., “PromptST: Prompt-enhanced spatio-temporal multi-attribute prediction,” in *Proc. ACM Int. Conf. Inf. Knowl. Manage.*, 2023, pp. 3195–3205.
[41] Z. Zhang, X. Zhao, H. Miao, C. Zhang, H. Zhao, and J. Zhang, “AutoSTL: Automated spatio-temporal multi-task learning,” in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 4902–4910.
[42] H. Yan, X. Ma, and Z. Pu, “Learning dynamic and hierarchical traffic spatiotemporal features with transformer,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22 386–22 399, Nov. 2022.
[43] G. Zerveas, S. Jayaraman, D. Patel, A. Bhamidipaty, and C. Eickhoff, “A transformer-based framework for multivariate time series representation learning,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2021, pp. 2114–2124.
[44] Y. Zhang and J. Yan, “Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting,” in *Proc. Int. Conf. Learn. Representations*, 2023.
[45] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *Proc. Int. Conf. Learn. Representations*, 2023.
[46] A. Vaswani et al., “Attention is all you need,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
[47] A. P. Mathur and N. O. Tippenhauer, “SWaT: A water treatment testbed for research and training on ICS security,” in *Proc. Int. Workshop Cyber-phys. Syst. Smart Water Netw.*, 2016, pp. 31–36.
[48] A. Abdulaal, Z. Liu, and T. Lancewicki, “Practical approach to asynchronous multivariate time series anomaly detection and localization,” in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2021, pp. 2485–2494.
[49] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. IEEE Int. Conf. Data Mining*, 2008, pp. 413–422.
[50] H. Xu et al., “Unsupervised anomaly detection via variational auto-encoder for seasonal KPIs in web applications,” in *Proc. World Wide Web Conf.*, 2018, pp. 187–196.
[51] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. Int. Conf. Learn. Representations*, 2015.
[52] A. Paszke et al., “PyTorch: An imperative style, high-performance deep learning library,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.

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